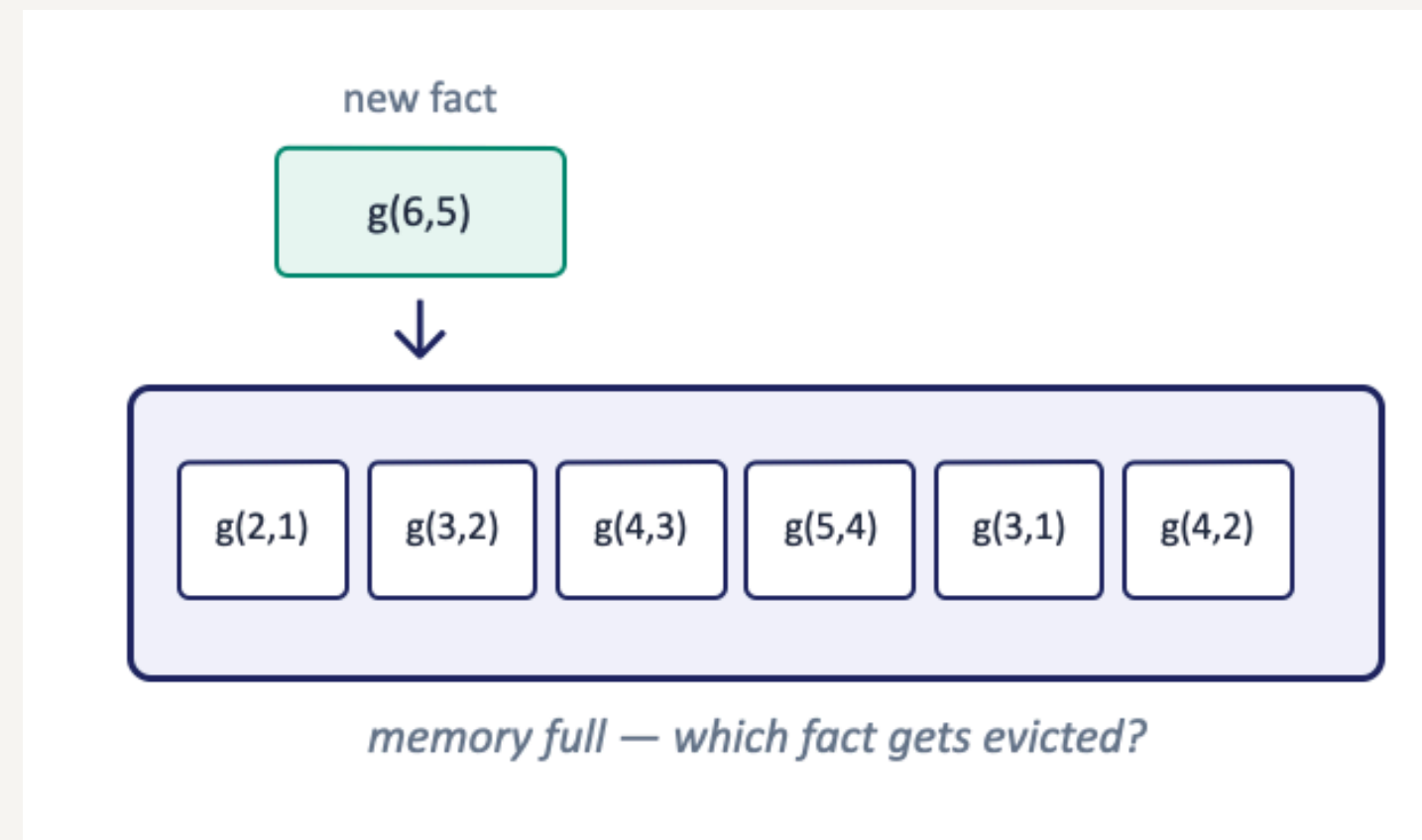

Limited Memory Inductive Reasoning Agent

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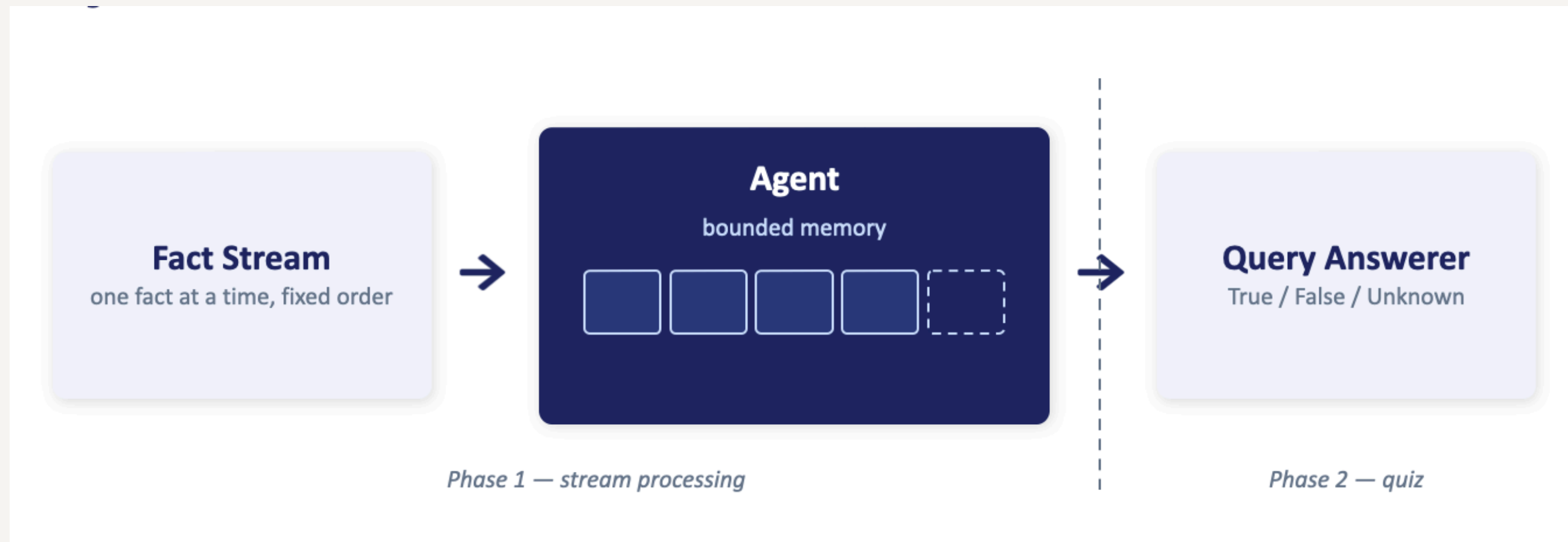


Project Idea: The Bounded Memory Problem



- Facts arrive one at a time from a stream, memory has a hard cap
- The agent must decide what to keep, what to forget, and what to generalize
- Naive strategies only pick which facts to drop, every dropped fact is lost forever
- **Goal:** Apply CSC 480 FOL and Inductive Reasoning topics to build an FOL compression agent that answers more queries correctly than naive agents

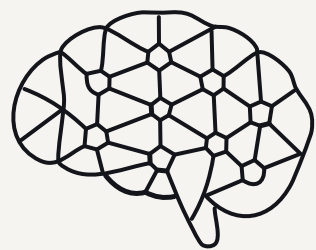
System Architecture



All 4 agents share the same *process(fact)* and *answer(query)*.

Four Agents, One Memory Limit

Agent	Eviction strategy	Uses rules?
SimpleAgent (Recent)	FIFO — evicts the oldest fact	No
RandomAgent	Evicts a random fact	No
ImportanceAgent	Evicts lowest-scored fact (frequency + shared args)	No
FOLCompressionAgent (ours)	Commits a general rule, evicts facts the rule can derive	Yes



FOL is the only agent that compresses. The others merely select.

Chain Datasets Force Inference

What the stream gives

greater: $2 \rightarrow 1$ $3 \rightarrow 2$ $4 \rightarrow 3$ $5 \rightarrow 4$...

divisibility: $2 \rightarrow 4$ $4 \rightarrow 8$ $8 \rightarrow 16$ $16 \rightarrow 32$...

mixed: both of the above interleaved
 $2 \rightarrow 1$ $2 \rightarrow 4$ $3 \rightarrow 2$ $4 \rightarrow 8$ $5 \rightarrow 4$ $8 \rightarrow 16$...

Only adjacent pairs are fed to the agent

What the quiz asks

greater: $\text{greater}(5, 1)?$ $\leftarrow$ skips 3 steps

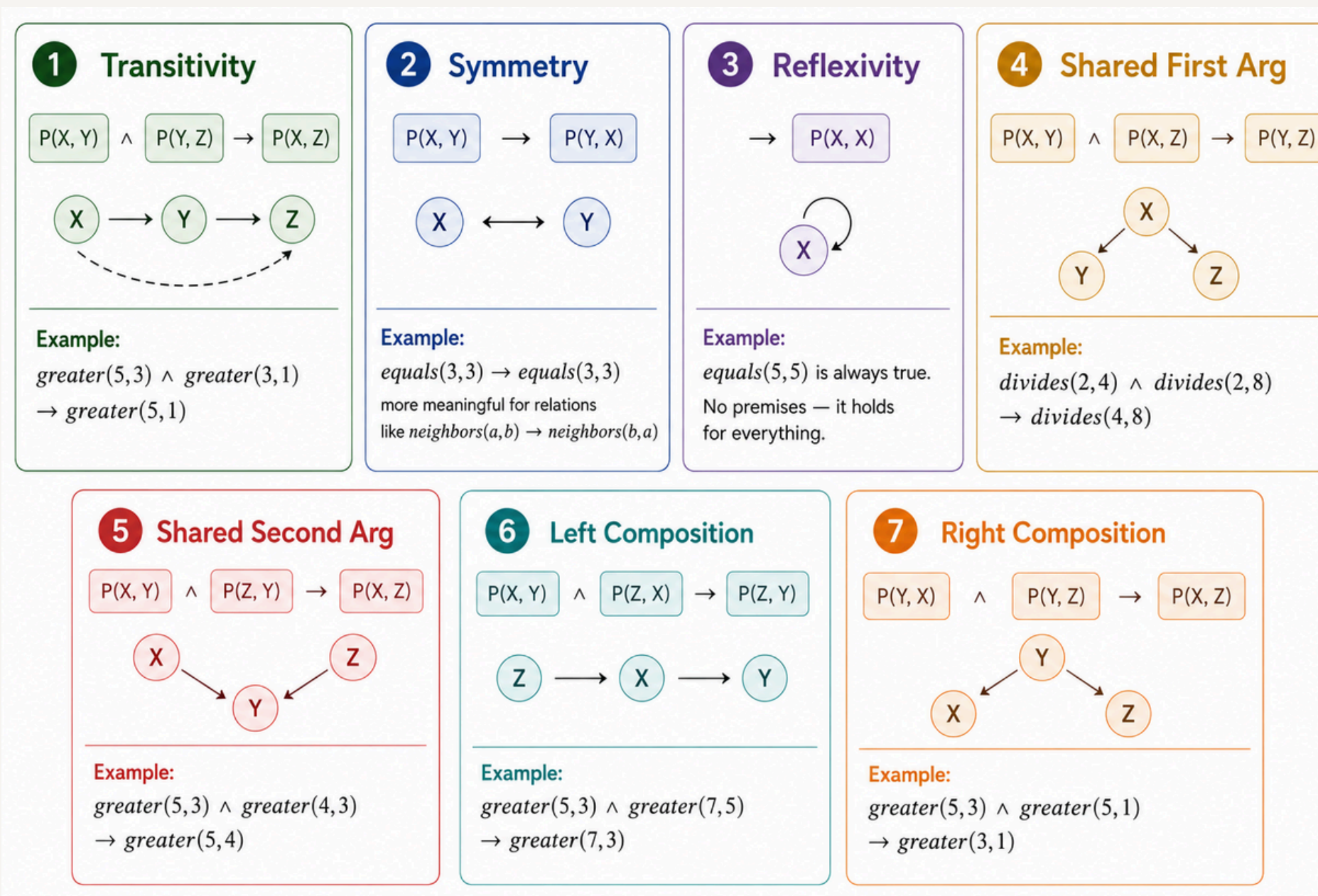
divisibility: $\text{divides}(2, 32)?$ $\leftarrow$ skips 3 steps

Quiz tests non-adjacent pairs
Requires transitivity to answer

- **Chain streams:** If you fed all possible pairs, every agent could answer by recall alone. Chains force inference to matter.
- **Fixed JSON files :** Quiz questions and stream order are generated once and saved. All agents see identical inputs so accuracy scores are directly comparable.

Seven Candidate Rule Templates

For any binary predicate p , the agent considers...



FOL Agent Pipeline

Five steps run on every incoming fact - *process(fact)*:



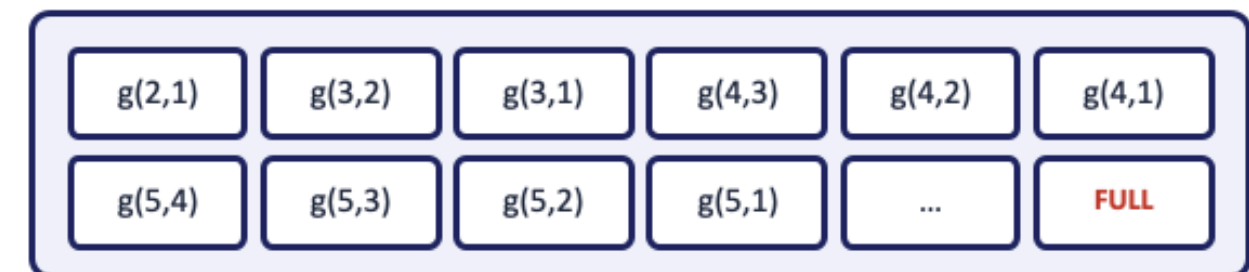
Answering Queries: memory facts + committed rules $\rightarrow$ Z3
entailment check $\rightarrow$ True/ False/ Unkown. Solver is cached,
rebuilt when memory changes.

★ This is the compression piece

Walkthrough: Learning Transitivity

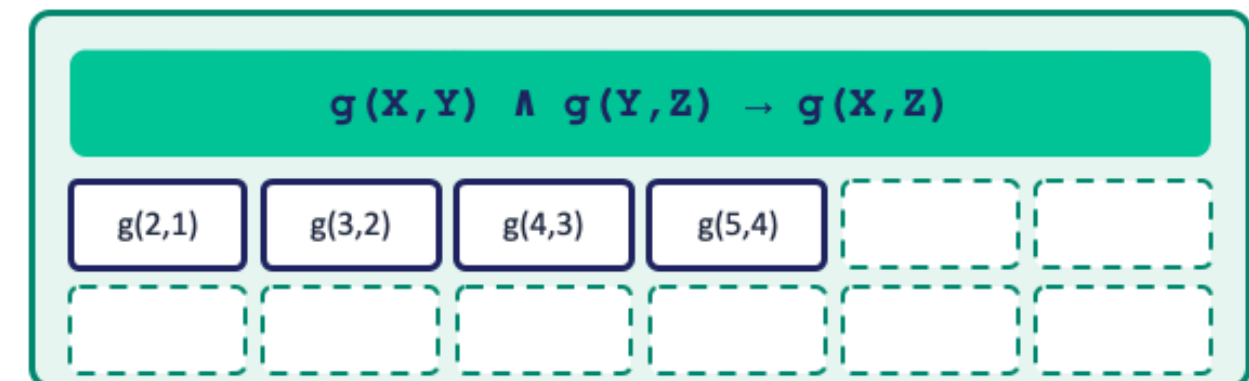
1. Stream delivers $\text{greater}(3,2)$, then $\text{greater}(2,1)$
2. 2nd arg of $g(3,2) = 1\text{st arg of } g(2,1) \rightarrow \text{pattern } (1,0) \rightarrow \text{transitivity } +1 \text{ vote}$
3. More adjacent pairs arrive, support reaches 3+
4. Support plateaus for 2 facts $\rightarrow$ rule commits
5. Facts entailed by other facts + the rule are evicted, e.g. $g(3,1)$ from $g(3,2) + g(2,1)$
6. Quiz asks $\text{greater}(5,1)$? $\rightarrow$ Z3 chains through the rule $\rightarrow$ True

Before commit — memory full of facts



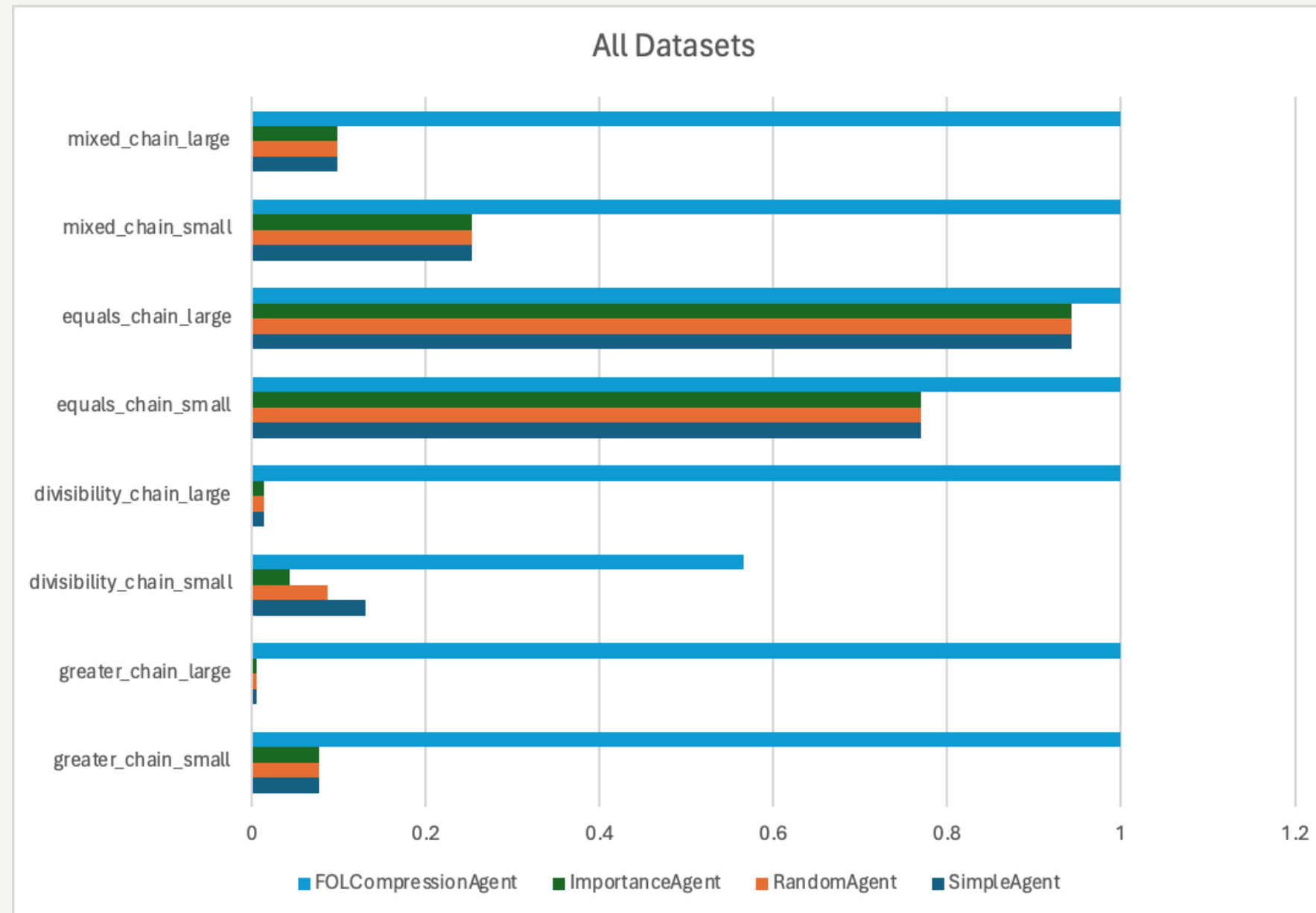
$\downarrow$ commit transitivity, evict derivable facts

After commit — rule + chain links + free slots



dashed = freed slots, ready for more chain links

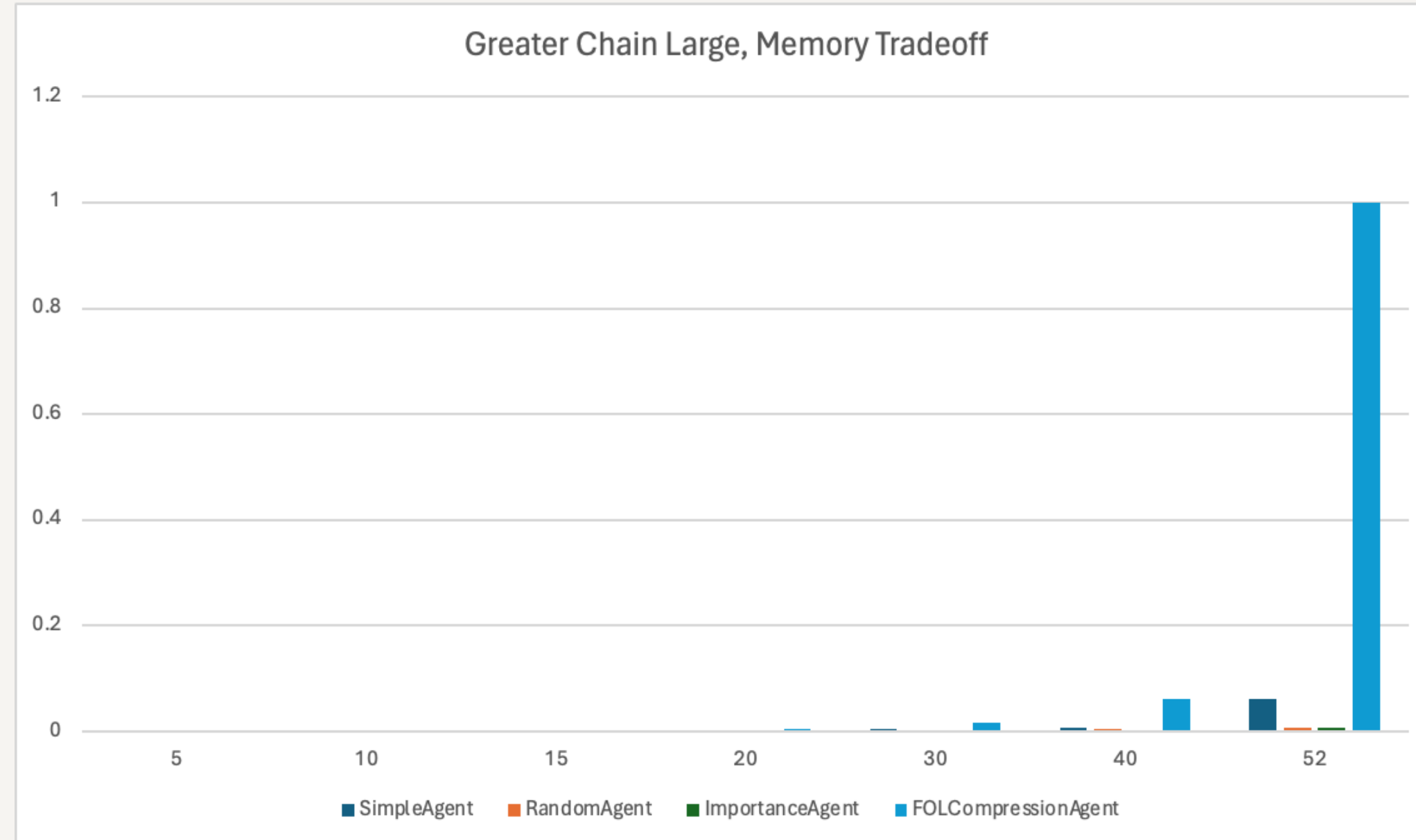
Results: FOL Dominates



100% on 7 of 8 datasets. Partial result: divisibility small (56.5%).

Short stream → too few fact pairs to reach 3 support votes before memory fills.

Memory Sweep: The Cliff



← rule commits ≈ here

Below the threshold, FOL behaves like SimpleAgent. The learning trace evicts pairs before support reaches 3. Naive agents flatline: more recall, still no inference.

The Headline Comparisons



- Simple, Random, and Importance score nearly identically everywhere; eviction strategy is irrelevant without rules.
- **The exception:** equals_chain, where naive agents reach 76.9–94.3%. Reflexivity facts make that quiz mostly recall.

Conclusions

Goal Achieved

- FOL compression beats naive limited-memory agents whenever inference is possible
- Rules preserve information that eviction destroys.

Known Limitation

- Cliff behavior at low memory
- Agent needs enough memory to see the pattern before it can compress it.

Future Work

- Softer / adaptive commitment thresholds
- Multi-predicate rules
- Learned support thresholds

“The best thing to forget is anything you can re-derive.”